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Biostatistics Project

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Goal. Test whether two image populations come from the same distribution.

Main questions

- Which representation should be used: raw pixels, CLIP, DINOv2, or a trained CNN?
- Which testing layer should be used: the original MMD test or lrj 's stabilized variant?
- Can we obtain high power while controlling Type-I error?

Representative experiments

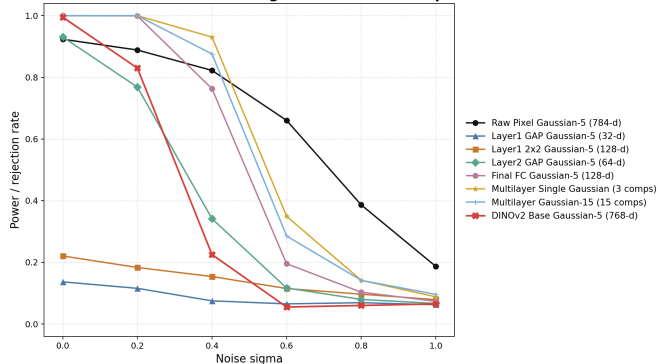
- MNIST: small-image sanity check
- BloodMNIST-224: biomedical cell images with noise
- PathMNIST-28: comparison with dechao's CNN experiments
- NCT-CRC: pathology images, sample size, and lrj comparison

Image data	Representation layer	Testing layer	Metrics
MNIST	Raw pixels	GAUSS5 / GEXP-5	Power
BloodMNIST	CLIP	NEW-MMMD	Type-I error
PathMNIST	DINOv2		sample-size
NCT-CRC	CNN features		curve

Implementation choice

Images are first converted into reusable `embeddings.npy` + `metadata.csv` pools. Testing then repeatedly samples from these pools. Type-I experiments use matched null splits or shared-pool designs to avoid false-positive inflation.

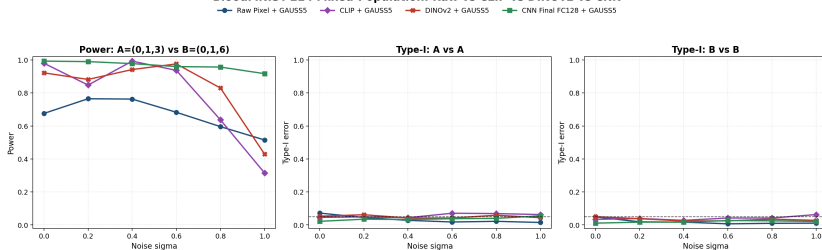
MNIST DINOv2 vs CNN Embedding MMD Power Comparison



σ	DINOv2 power
0.0	0.995
0.2	0.830
0.4	0.225
0.6	0.055
0.8	0.060
1.0	0.065

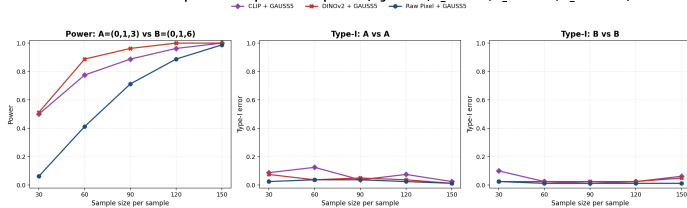
Interpretation. MNIST images are 28×28 grayscale digits. DINOv2 requires resizing to a much larger input size, and the combination of resizing, interpolation, and Gaussian noise creates a strong domain mismatch.

BloodMNIST-224 Mixed-Population: Raw vs CLIP vs DINOv2 vs CNN



Method	Low-noise power	High-noise power	Type-I behavior
Raw pixel + GAUSS5	moderate	retains some signal	conservative
CLIP + GAUSS5	strong at low noise	drops at $\sigma = 1.0$	near nominal
DINOv2 + GAUSS5	strong at medium noise	better than CLIP	near nominal
CNN final + GAUSS5	strongest overall	0.917 at $\sigma = 1.0$	slightly conservative

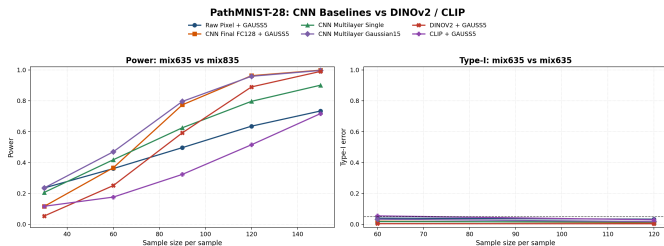
BloodMNIST Mixed-Population Sample-Size Comparison ($\sigma=0.6$, $n_{\text{outer}}=4$, $n_{\text{inner}}=20$, $B_{\text{boot}}=100$)



Fixed noise level: $\sigma = 0.6$.

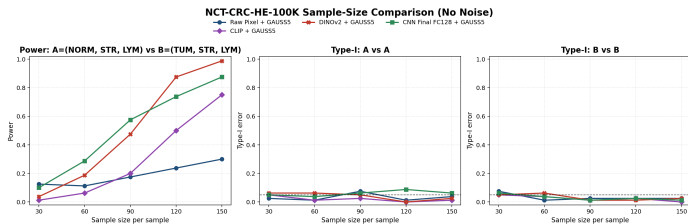
n	Raw	CLIP	DINOv2
30	0.063	0.500	0.513
60	0.413	0.775	0.888
90	0.713	0.888	0.963
120	0.888	0.963	1.000
150	0.988	1.000	1.000

Conclusion. Learned embeddings greatly improve sample efficiency. Raw pixels need substantially larger samples to catch up.



n	CLIP	DINOv2	Note
30	0.117	0.053	both weak
60	0.176	0.251	DINOv2 improves
90	0.323	0.593	gap increases
120	0.515	0.890	DINOv2 strong
150	0.717	0.990	near full power

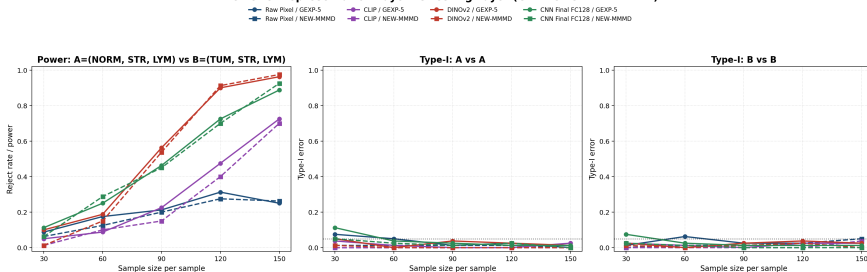
Conclusion. DINOv2 becomes strong as sample size grows, but task-trained CNN baselines remain important in the small-to-medium sample regime.



n	Raw	CLIP	CNN	DINOv2
30	0.125	0.013	0.100	0.038
60	0.113	0.063	0.288	0.188
90	0.175	0.200	0.575	0.475
120	0.238	0.500	0.738	0.875
150	0.300	0.750	0.875	0.988

Conclusion. CNN is stronger at small and medium sample sizes, while DINOv2 becomes the strongest representation at larger sample sizes.

NCT-CRC: Representation Layer vs Testing Layer (GEXP-5 vs NEW-MMMD)



- **GEXP-5:** original multi-kernel MMMD-style testing layer; often has higher small-sample power.
- **NEW-MMMD:** lrj's covariance-stabilized variant with kernel pruning; often lowers Type-I error.
- At $n = 150$, NEW-MMMD matches or slightly exceeds GEXP-5 for DINOv2 and CNN features, but it can lose power in small samples.

Main findings

- The representation layer is the dominant factor: raw pixels, CLIP, DINOv2, and CNN features lead to very different testing power.
- On BloodMNIST-224, the task-trained CNN is the most robust representation under Gaussian noise.
- On NCT-CRC, DINOv2 overtakes the CNN at large sample sizes, while the CNN is more competitive at smaller sample sizes.
- CLIP is usually weaker than DINOv2 for these morphology-driven biomedical tasks.
- Irj's NEW-MMMD is best viewed as a stabilization and Type-I-control improvement, not a guaranteed power booster.

Next steps

Increase Monte Carlo repeats, run NEW-MMMD on BloodMNIST, add stain/color perturbations, and compare with stronger biomedical encoders.